

EMRE BATU ATAKAN

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EDUCATION

University of Waterloo

Waterloo, Ontario, Canada | September 2025 - April 2029

Bachelor of Mathematics in Computer Science (Honours) with Co-op, GPA: 3.7

- Foundations of Sequential Programming, Object-Oriented Software Development, Designing Functional Programs, Calculus 2, Linear Algebra 1

Awards

- **ChessHacks Waterloo 2025:** Pufferfish Chess Engine, **3rd Place**
- **American Mathematics Olympiad 2024-2025:** Silver Medal (National 7th / Global 269th) in 2024; Bronze Medal in 2025
- **International STEM Olympiad:** 2 x Mathematics Gold Medal; Global 12th in 2024, Global 14th in 2023
- **UWaterloo Math Contests:** Fermat 2024 Honor Roll, Distinction; Cayley 2023 Distinction

EXPERIENCE

University of Waterloo | Robotics ML Research Lead

Waterloo, Ontario, Canada | June 2026 - Present

- Co-lead an 8-person team developing DINOv3-Transformer, pretrained ConvNeXt, and CNN/MLP visuomotor behavior-cloning policies.
- Built reproducible **A100/H100 training infrastructure** with PyTorch, Slurm, EMA checkpoints, and resumable jobs; increased training throughput **9x** by caching vision-encoder embeddings.
- Ran hyperparameter sweeps across temporal horizons, auxiliary losses, and dataset mixtures to isolate the drivers of policy performance.
- Diagnosed Transformer overfitting by comparing offline training loss against closed-loop performance, then developed and trained a smaller CNN/MLP policy that improved randomized pick-and-place success from **30% to 99.8%**, with only **4 failures across 2,000+ seeded rollouts**.
- Built and validated an end-to-end MuJoCo/PyTorch training and evaluation stack spanning data collection, model training, W&B experiment tracking, dashboards, logging, and reproducible closed-loop benchmarking.
- Developed an asynchronous closed-loop inference (model-sim-control) on MuJoCo with 100 Hz command execution and 10 Hz inference rate.

Shopify Inc. | Software Engineering Intern

Toronto, Ontario, Canada | Winter 2026

Automated Shop Analysis Pipeline

- Built a distributed shop-analysis pipeline from scratch: incremental selection and storage (dbt, BigQuery), distributed ML-service execution (Apache Flink, custom Java UDF, FastAPI), and daily orchestration (Airflow), processing **100K+ shops/day across an 800K+ corpus**.
- Ported the prototype into Shopify's monorepo, developing three specialized AI agents for fraud detection, business profiling, and merchant classification, along with a parallel orchestrator and **129 unit tests**.
- Implemented an aiohttp pre-filter issuing **500 concurrent requests** to skip inaccessible sites before GPU inference.
- Built benchmarks tracking throughput, success rates, and P50/P95/P99 latency to tune stage concurrency.

River AI Agent Evaluation Framework and Analytics Dashboard

- Built a reproducible A/B evaluation framework for River, Shopify's company-wide Slack-native AI coding agent co-authoring **1 in 8 merged PRs**; compared skill configurations on identical prompts with LLM judges and deterministic assertions, publishing results to BigQuery.
- Built a dashboard to analyze River adoption, usage by channel and time, and recurring tool failures (BigQuery)
- Fixed authentication and publishing failures blocking evaluations in production containers.

Thresh Power | Software Development Intern

Summer 2024

- Built and deployed a GCP-hosted Python microservice using the OpenAI Assistants API to generate due-diligence reports from user files; established staging and production environments and developed LLM benchmarks to evaluate prompting strategies.
- Shipped a client-facing streaming file-download feature (React, Python) with **100% API unit test coverage**, supporting daily use.

PROJECTS

Pufferfish | Chess Engine

2025 - 2026

- Built a Python NNUE engine and C++ prototype with iterative-deepening alpha-beta search; trained PyTorch evaluators on up to **155M positions** using Modal A100s, reaching 1,642 simulated Elo and placing **3rd of 75 teams** at ChessHacks Waterloo.

Speech to Phonemes

Summer 2025

- Built a multiprocessing pipeline to create a **120-hour speech-to-phoneme dataset** using batched Whisper GPU inference, multithreaded OpenAI generation to add **4,000 out-of-vocabulary pronunciations** to CMUDict, and dynamic-programming alignment.

LEADERSHIP

FIRST Robotics Competition | Team Captain

2023 - 2025

Co-founded RedFox 10247 and captained two FRC teams, leading up to 122 members, raising \$20K+, training 23 newcomers through Java robot-control simulations and electronics/mechanical workshops, and building a CNC-equipped workshop; led RedFox to 11th in its regional debut.

SKILLS

Languages: Python, C++, C, Java

ML & Robotics: PyTorch, MuJoCo, ROS, W&B, Whisper, SolidWorks

Systems & Data: Slurm, Docker, FastAPI, Apache Flink, Airflow, dbt, BigQuery, GCP